

Mohammad Maaz Rashid

New York, USA — mr7374@nyu.edu — (929) 726-4505 — linkedin.com/in/rashid-maaz

EDUCATION

New York University

M.S. in Computer Science (GPA 3.78/4)

• Coursework: Design and Analysis of Algorithms, Foundations of Data Science, Machine Learning, Deep Learning, Big Data

New York, USA

Sept 2024 – May 2026

SRM Institute of Science and Technology

BTech in Computer Science and Engineering (CGPA 9.31/10)

Chennai, India

May 2022

SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++, Bash
- **Frameworks:** Claude (Anthropic), LangChain, Hugging Face (BERT, T5), GPT-3, LLaMA, TensorFlow, PyTorch, Scikit-Learn
- **Cloud & Platform Tools:** AWS (SageMaker, Lambda, EC2, ECS, Glue, EMR, Batch, Athena, Step Functions, Kinesis, Textract, EKS, IAM, Cognito, ALB, Route 53, Secrets Manager, CloudWatch, CloudFront, CloudFormation), Docker, FastAPI
- **Datastores:** S3, Redshift, DynamoDB, Elasticsearch, Neo4j, ElastiCache, AWS RDS

EXPERIENCE

Qualcomm

Machine Learning Intern

San Diego, USA

May 2025 – Aug 2025

- Applied **Retrieval-Augmented Generation (RAG)** and **DSL-based prompt engineering** with **Anthropic's Claude** to streamline language-based validation of modern framework test scenarios, and fine-tuned **Alibaba's Qwen** model to enhance **natural language understanding**, accelerating internal QA and validation workflows.
- Engineered a **JSON-based knowledge repository** by programmatically extracting handler keyword definitions and documentation from the **MTFpy codebase**, and enriched it with real-world parameterized examples mined from **20GB+ of SCC logs** for future **LLM fine-tuning** and retrieval workflows.
- Built scalable, distributed **data processing pipelines** leveraging **AWS (S3, Lambda, Glue, EMR)** and **Apache Hadoop** to ingest, parse, and transform **multi-terabyte** daily logs and telemetry data; ensured fault tolerance and efficient batch processing using **HDFS** and **MapReduce**.
- Developed and deployed **cloud-native analytics modules** using **Spark on EMR** to extract **real-time insights** from high-throughput data streams, integrating structured output into **Athena** and **Redshift** for downstream analytics.

Barclays

Software Engineer

Pune, India

Aug 2022 – June 2024

- Automated internal business processes, reducing manual workload by **45%**, through the development and integration of **LLM-powered features** such as **intelligent chatbots**, **entity extraction**, **image generation**, **document search**, and **text summarization**.
- Enhanced contextual understanding by fine-tuning **Hugging Face's Instructor XL (BART)** model to improve entity extraction from complex internal documents.
- Integrated **Chroma Vector Database** for efficient storage and retrieval of **vector embeddings**, significantly improving data processing accuracy and operational efficiency. *The business impact of this implementation was estimated to be worth "eight figures", as noted in an executive appreciation email.*
- Preprocessed and managed over **5 TB** of financial data daily using **AWS (S3, EMR, Glue)**, ensuring scalable and reliable **distributed data pipelines**.
- Achieved a **60% reduction** in data processing time, significantly improving business efficiency, reducing operational costs, and decreasing manual errors by **20%**.
- Streamlined **receivables and payments processing**, resulting in a **15% reduction** in processing time.

PROJECTS

Focus Assist: AI-powered Pomodoro Tracker

- Built a thread-safe **Pomodoro timer** in **Python** for **Snapdragon X Elite** laptops, featuring CLI-based task tracking, customizable intervals, and **TUI progress bars**. Designed for future on-device AI using **NPU inference** and **local model integration**, enabling real-time focus tracking with efficient edge compute utilization.

ResNet Architecture Optimization

- Optimized a ResNet based deep learning architecture by strategically tuning channel dimensions, convolutional filter sizes, and skip connections, achieving superior test accuracy with fewer than **5 million parameters**, while also enhancing **ResNet-50 to reach 87% efficiency** in image classification tasks.

AWARDS

Qualcomm's HaQathon - Winner

- Developed and deployed Focus Assist, a Pomodoro timer using on-device Snapdragon X Elite NPU in day-night hackathon.

Barclays World Class Customer Service Award

- Demonstrated dedication to issue resolution, collaboration with relevant teams, and commitment to delivering world-class service aligned with Barclays' standards.

C Programming - Expert level

- Recognized for exceptional performance as the top student among 600 peers throughout the year by Centre for Applied Research in Education — SRM Institute of Science and Technology.